

Digital Self-Efficacy Mediation in Digital Leadership: A PLS-SEM Analysis of ICT Preparedness and Employee-AI Support

Abstract

This study examines how Employer Preparedness for ICT and AI Support influence Sustainable Digital Leadership within the Indian IT sector, with Digital Self-Efficacy tested as a mediating mechanism. As digital transformation reshapes the nature of leadership in technology intensive industries, leadership can no longer be evaluated on technical competence alone; it must also reflect confidence, responsibility, and long-term sustainable orientation. Despite growing interest in these themes, empirical work connecting them within large Indian IT organizations remains limited. Data were collected from 384 employees across five major IT firms Infosys, TCS, Wipro, Cognizant, and Accenture India using a validated Likert scale questionnaire, and analyzed using Partial Least Squares Structural Equation Modelling (PLSSEM) in SmartPLS. Results show that both ICT preparedness and AI Support significantly influence Sustainable Digital Leadership, both directly and indirectly through Digital Self-Efficacy. AI Support emerged as the strongest overall driver of leadership outcomes, while ICT preparedness was the strongest predictor of Self-Efficacy. All seven hypotheses were supported, confirming Digital Self-Efficacy as a critical psychological bridge between organizational digital enablers and sustainable leadership behavior.

Keywords: Sustainable Digital Leadership, Digital Self-Efficacy, Employer Preparedness for ICT, AI Support, PLSSEM, Indian IT Sector

Introduction

The Indian IT industry has evolved from a modest software services base in the early 1990s into one of the world's largest and most influential technology ecosystems, anchored in hubs such as Bengaluru, Hyderabad, Chennai, Pune, and Mumbai. Growth has been driven

by rising global demand for digitalization, the rapid adoption of cloud computing, and the accelerating integration of artificial intelligence into everyday business processes. This transformation has fundamentally changed what is expected of leadership within the sector. Leaders are no longer judged solely on technical delivery; they are increasingly expected to guide organizations through digital complexity in ways that are ethical, people centered, and sustainable over the long term.

Despite heavy investment in AI tools and ICT infrastructure across major Indian IT firms, this investment does not automatically translate into strong leadership outcomes. Many organizations adopt advanced technology without fully understanding how organizational preparedness and AI Support interact with individual employee confidence to shape leadership behavior. Employees vary considerably in their belief in their own ability to use digital tools effectively, and this variation can determine whether technology investment translates into genuine leadership capability or remains underutilized infrastructure.

Although digital leadership, AI adoption, and ICT readiness have each been studied independently, little empirical research has combined these constructs into a single mediation framework, particularly within the Indian IT context. This study addresses that gap by examining whether Digital Self-Efficacy mediates the relationship between two organizational enablers Employer Preparedness for ICT and AI Support and Sustainable Digital Leadership.

Objectives

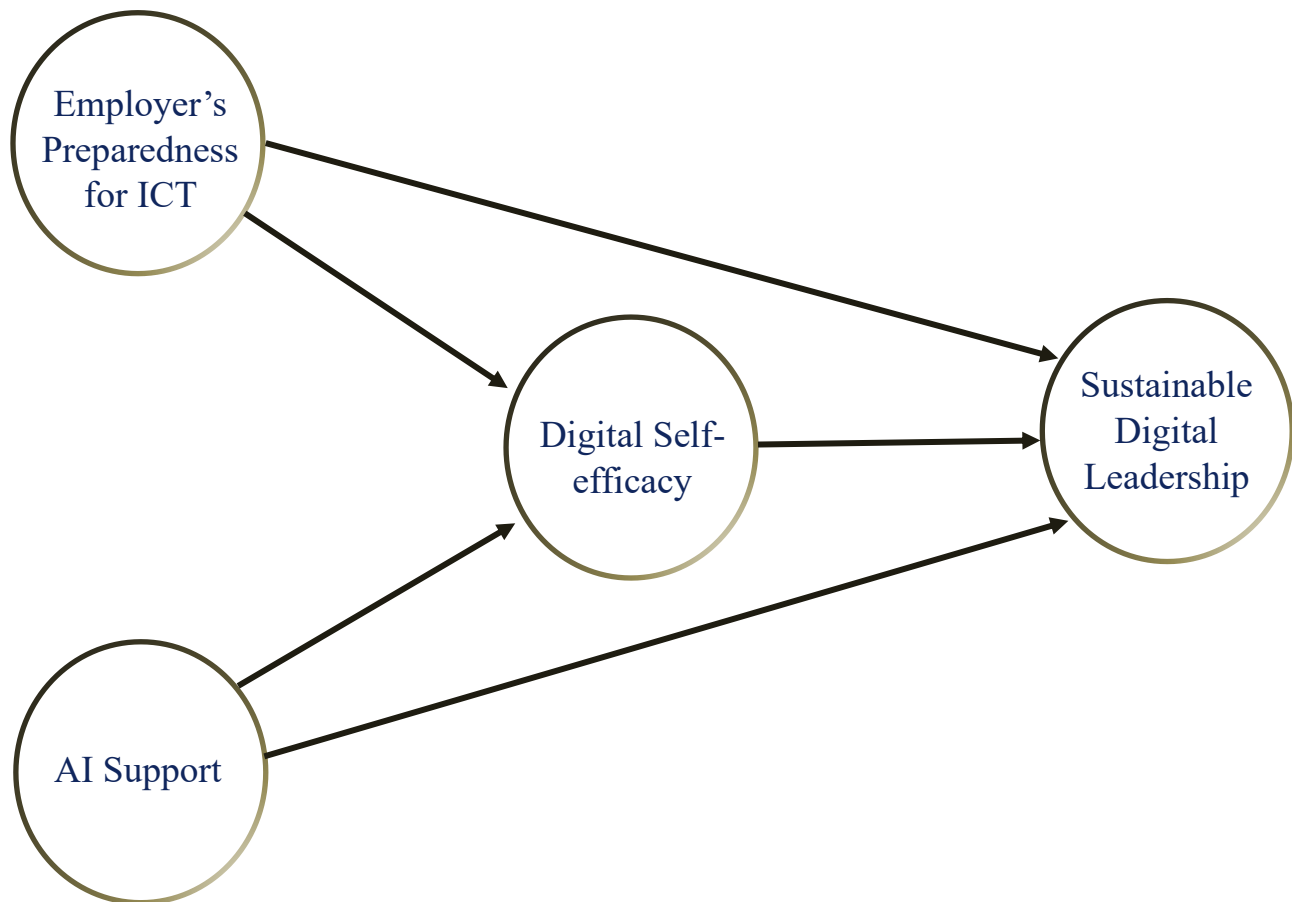
- To analyze the effects of Employer Preparedness for ICT, AI Support, and Digital Self-Efficacy on Sustainable Digital Leadership.
- To examine the mediating role of Digital Self-Efficacy in the relationship between Employer Preparedness for ICT, AI Support, and Sustainable Digital Leadership.

Scope of the Study

The study focuses on five leading technology organizations Infosys, TCS, Wipro, Cognizant, and Accenture India selected for their scale, global presence, and active

engagement with digital transformation and AI integration. Employees across various hierarchical levels within these firms formed the study population. While the findings offer meaningful insight for technology driven organizations, they may not generalize directly to manufacturing, public sector, or non-IT contexts, where organizational dynamics differ considerably.

Conceptual model



Literature Review and Theoretical Framework

Theoretical Foundation

This study is grounded in Bandura's Social Cognitive Theory, particularly his concept of Self-Efficacy, defined as an individual's belief in their own capacity to execute tasks and

achieve desired outcomes (**Bandura, 1986**). Self-Efficacy is shaped by mastery experiences, vicarious learning, social persuasion, and physiological states, and functions as a mediating mechanism between environmental conditions and individual behavior (**Bandura, 1977**). Applied to this study, organizational investment in ICT preparedness and AI Support creates conditions through training, technological exposure, and institutional encouragement that strengthen employees' belief in their digital capabilities, providing theoretical justification for positioning Digital Self-Efficacy as the study's mediating variable.

Sustainable Digital Leadership

Digital leadership was found positively associated with innovation management in the telecom sector, though its narrow industrial scope constrained broader applicability (**Wasono & Furinto, 2018**). Bibliometric mapping of digital leadership literature revealed persistent fragmentation across disciplinary silos, with insufficient integration between strategic leadership and technology management traditions (**Karakose et al., 2022**). Digital leadership capability was demonstrated to drive innovation performance through platform digitization as a mediating pathway, establishing that infrastructure rather than leadership behavior alone is the proximate performance driver (**Benitez et al., 2022**). Digital transformation was established as a mediating mechanism between digital leadership and firm performance, confirming that leadership must manifest through transformative organizational processes to yield measurable outcomes (**Senadjki et al., 2024**). Digital culture and employee digital capabilities were found to fully mediate the digital leadership organizational performance relationship, underscoring that leadership effectiveness is codetermined by the organizational ecosystem surrounding it (**Shin et al., 2023**).

Digital Self-Efficacy

Employee Self-Efficacy was identified as an antecedent of digital transformation but was subsumed within a transformational leadership framework rather than examined as a discrete construct (**Gun et al., 2024**). ICT Self-Efficacy was demonstrated to vary significantly across generational cohorts in digitally transforming organizations,

complicating aggregated singleconstruct measurement approaches (**Komarkova & Egerova, 2023**). Technical Self-Efficacy was shown to mediate the relationship between innovative digital leadership and employee performance, positioning digital confidence as the operative cognitive mechanism linking leadership to output (**Elsawy & AbuAlhaija, 2026**). Digital Self-Efficacy was further demonstrated to moderate the relationship between ICT exposure and job dissatisfaction, such that digitally confident employees transform technological demand into productive engagement (**Busse et al., 2022**). Digitally selfefficacious employees were found to exhibit significantly greater adaptability and agility in highdemand digital workplaces, confirming DSE as a reliable predictor of productive digital functioning (**Maran et al., 2022**).

Employer Preparedness for ICT

ICT and R&D were found to produce differential effects on total factor productivity growth at the country level, undermining technologically deterministic policy prescriptions (**Edquist & Henrekson, 2017**). Workrelated ICT use outside office hours was found to significantly increase worktohome conflict, cautioning against treating ICT deployment as an unambiguously enabling organizational intervention (**Gadeyne et al., 2018**). ICT was demonstrated to drive organizational performance through the catalyzing mediation of corporate entrepreneurship and innovation capabilities, establishing that entrepreneurial culture is prerequisite to productive ICT investment (**Yunis et al., 2018**). Workforce skills, regional proximity, and startup orientation were found to jointly moderate the ICTinnovation relationship, reinforcing human capital as an essential complement to technological infrastructure (**Alam et al., 2022**). ICT adoption decisions among SMEs were found to be structurally conditioned by firm characteristics and institutional context, confirming the contingent nature of ICT preparedness as an organizational variable (**Giotopoulos et al., 2017**).

AI Support

Technological, organizational, and environmental factors shaping AI adoption in manufacturing were identified through a TAMTOE model, yet the performance and behavioral consequences of postadoption AI Support remained largely unaddressed

(Chatterjee et al., 2021). A goal programming model for humanAI team formation was proposed, acknowledging that optimizing such arrangements is nontrivial, yet the contribution remained at a remove from the empirical realities of how AI Support shapes employee psychological resources (La Torre et al., 2023). AIemployee collaboration was demonstrated to significantly enhance business performance through KnowledgeBased View and Sociotechnical Systems frameworks, contingent upon knowledgesharing norms and socialization structures that preserve human expertise alongside AI capabilities (Chowdhury et al., 2022). AI Support was found to enhance employees' proactive behavior by augmenting cognitive resources and reducing attentional load, establishing a clear psychological pathway through which AI support translates into initiativetaking behavior (Sun et al., 2025). AIemployee collaboration was found to increase counterproductive work behaviors unless leader emotional support serves as a moderating buffer, indicating that AI integration without accompanying managerial infrastructure risks relational disruption (Meng et al., 2025). AI Support was further confirmed to improve efficiency and job satisfaction in public administration when organizational support structures accompany deployment, reinforcing that contextual conditions determine the quality of AI Support outcomes (Rulandari & Silalahi, 2025).

Research Gap

Existing research has largely examined digital leadership, digital Self-Efficacy, ICT readiness, and AI adoption as separate streams, with limited attention to how they interact within a single mediation framework. Digital Self-Efficacy has been studied in the context of technology adoption, but its mediating role between organizational digital readiness and sustainable leadership outcomes remains underexplored, and AI Support has largely been examined from a productivity standpoint rather than its influence on longterm leadership capability. Empirical evidence from the Indian IT sector an industry undergoing rapid AI integration remains particularly limited, a gap this study addresses.

Hypotheses

H1: Employer Preparedness for ICT positively influences Sustainable Digital Leadership.

H2: AI Support positively influences Sustainable Digital Leadership.

H3: Employer Preparedness for ICT positively influences Digital Self-Efficacy

H4: AI Support positively influences Digital Self-Efficacy

H5: Digital Self-Efficacy positively influences Sustainable Digital Leadership.

H6: Digital Self-Efficacy mediates the relationship between Employer Preparedness for ICT and Sustainable Digital Leadership.

H7: Digital Self-Efficacy mediates the relationship between AI Support and Sustainable Digital Leadership.

Research Methodology

Research Design

The study follows a quantitative, cross-sectional survey design, collecting primary data from employees across the five selected organizations using a structured questionnaire built on validated scales drawn from Scopus indexed publications.

Sample and Sampling Method

A stratified random sampling approach was used to ensure representation across firms and employee categories (**technical and nontechnical roles**). Sample size was calculated using Cochran's formula at a 95% confidence level and 5% margin of error, yielding a minimum requirement of approximately 384 respondents (**Cochran, 1977**).

Population and Sample Size was allocated by the proportional sample allocation across the five firms, with TCS contributing the largest share (120) given its comparatively larger workforce, and Cognizant the smallest (54). The allocation ensures that each organization

is represented in proportion to its actual employee strength, making the overall sample more reflective of the sector as a whole.

Instrument

The questionnaire comprised 37 items across four constructs, measured on a 5point Likert scale (**1 = Strongly Disagree to 5 = Strongly Agree**).

Questionnaire items

Sl	Constructs	Items	Source
1.	Sustainable Digital Leadership	5	(Uzunbacak, 2021)
2.	Digital Self-efficacy	12	(Paredes-aguirre et al., 2024)
3.	Employer's Preparedness For ICT	10	(Komarkova & Egerova, 2023)
4.	AI Support	10	(Divya et al., 2024)
	Total Items	37	

This table lists the number of items measuring each construct and the validated source scale each was adapted from, forming the basis of the 37item instrument. Using previously validated scales strengthens the credibility of the measurement approach, since these items have already been tested for reliability in earlier published research.

Demographic variables age, gender, work experience, and designation were included for descriptive analysis.

Analytical Tool

Data were analyzed using SmartPLS through Partial Least Squares Structural Equation Modelling (PLSSEM), chosen for its ability to simultaneously assess measurement and

structural paths, making it well suited for testing the mediating role of Digital Self-Efficacy (Hair et al., 2022).

Limitations

The study is restricted to five organizations, which, despite their scale, may not represent the full diversity of the Indian IT industry, including midsize and emerging firms. Reliance on self-reported data introduces potential social desirability bias and common method variance. The cross-sectional design limits causal inference, and geographical concentration in major IT hubs may not reflect the experience of professionals in smaller cities. Finally, the model captures a selective set of constructs and does not account for other variables that may influence sustainable digital leadership.

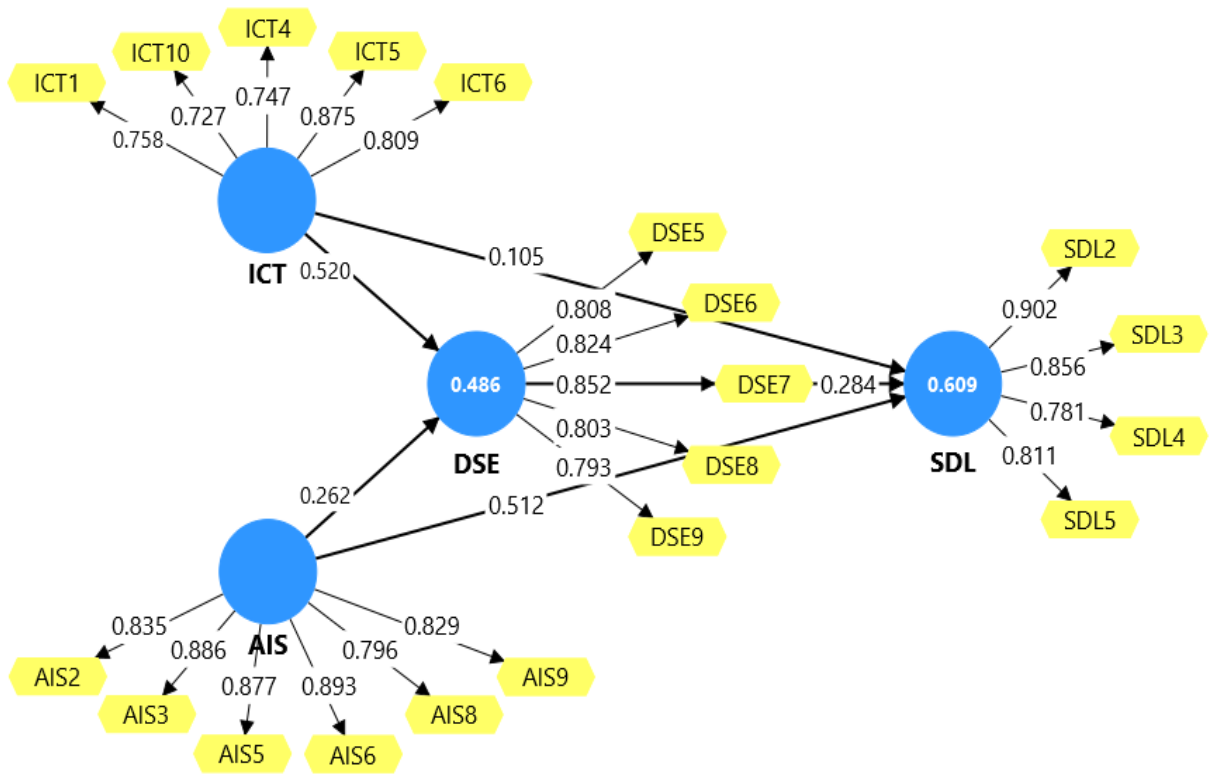
Data Analysis and Results

Demographic Profile

Respondent characteristics, showing a predominantly young, early-career sample concentrated in the 20–30 age range with 1–3 years of experience, and the majority choosing not to disclose gender. This profile suggests the findings may be more representative of junior and early-career employee perspectives than of senior leadership viewpoints.

Graphical Output:

PLS Algorithm



Measurement Model

Outer Loadings and Reliability

A few items initially fell below the acceptable loading value and were removed, which simply means those specific questions were not measuring their intended concept clearly enough and would have weakened the accuracy of the results. Once removed, all remaining items loaded strongly on their intended construct, and all four constructs exceeded the minimum reliability (**0.70**) and AVE (**0.50**) thresholds. This confirms that the final questionnaire items reliably and consistently measure what they were designed to measure.

Construct	Indicator	Outer loadings	Indicator Reliability	Composite reliability (rho_c)	Average variance extracted (AVE)
AIS (AI Support)	AIS2 <- AIS	0.835	0.697	0.941	0.728
	AIS3 <- AIS	0.886	0.785		
	AIS5 <- AIS	0.877	0.769		
	AIS6 <- AIS	0.893	0.797		
	AIS8 <- AIS	0.796	0.634		
	AIS9 <- AIS	0.829	0.687		
DSE (Digital Self-efficacy)	DSE5 <- DSE	0.808	0.653	0.909	0.666
	DSE6 <- DSE	0.824	0.679		
	DSE7 <- DSE	0.852	0.726		
	DSE8 <- DSE	0.803	0.645		
	DSE9 <- DSE	0.793	0.629		
ICT	ICT1 <- ICT	0.758	0.575	0.889	0.616
	ICT10 <- ICT	0.727	0.529		

(Employer preparedness for ICT)	ICT4 <- ICT	0.747	0.558		
	ICT5 <- ICT	0.875	0.766		
	ICT6 <- ICT	0.809	0.654		
SDL (Sustainable Digital Leadership)	SDL2 <- SDL	0.901	0.812	0.904	0.703
	SDL3 <- SDL	0.856	0.733		
	SDL4 <- SDL	0.781	0.610		
	SDL5 <- SDL	0.811	0.658		

Discriminant Validity

Heterotrait-Monotrait Ratio (HTMT)

	AIS	DSE	ICT	SDL
AIS				
DSE	0.596			
ICT	0.583	0.725		
SDL	0.787	0.731	0.643	

All HTMT values fell below the 0.90 threshold, confirming that the four constructs are empirically distinct from one another rather than overlapping. This means respondents were able to meaningfully differentiate between concepts like AI Support and ICT Preparedness when answering the survey, rather than treating them as the same idea.

FornellLarcker Criterion

	AIS	DSE	ICT	SDL
AIS	0.853			
DSE	0.542	0.816		
ICT	0.538	0.662	0.785	
SDL	0.723	0.631	0.568	0.839

The square root of AVE for each construct (diagonal values) exceeded all off-diagonal inter-construct correlations, confirming discriminant validity at the construct level. In simple terms, each construct shares more in common with its own items than with the items of any other construct, further supporting that the four variables are genuinely separate concepts.

CrossLoadings

	AIS	DSE	ICT	SDL
AIS2	0.835	0.323	0.377	0.639
AIS3	0.886	0.558	0.464	0.708
AIS5	0.877	0.503	0.499	0.620
AIS6	0.893	0.528	0.514	0.636
AIS8	0.796	0.429	0.410	0.531
AIS9	0.829	0.397	0.482	0.541
DSE5	0.599	0.808	0.394	0.521
DSE6	0.343	0.824	0.639	0.635
DSE7	0.498	0.852	0.682	0.477
DSE8	0.383	0.803	0.449	0.483
DSE9	0.397	0.793	0.490	0.440
ICT1	0.299	0.478	0.758	0.381
ICT10	0.353	0.408	0.727	0.368
ICT4	0.375	0.418	0.747	0.356
ICT5	0.604	0.736	0.875	0.604

ICT6	0.395	0.451	0.809	0.446
SDL2	0.731	0.472	0.585	0.901
SDL3	0.470	0.521	0.478	0.856
SDL4	0.432	0.681	0.475	0.781
SDL5	0.734	0.470	0.366	0.811

All retained items loaded most strongly on their intended construct and noticeably weaker on all others, confirming discriminant validity at the item level. This item level check reinforces the construct level results above, showing that the pattern of distinctiveness holds true even down to individual questionnaire items.

Structural Model

Collinearity Statistics (VIF)

Collinearity for Indicators: VIF Outer Model

	VIF
AIS2	3.243
AIS3	3.726
AIS5	3.359
AIS6	3.503
AIS8	2.146
AIS9	3.362
DSE5	2.074
DSE6	2.002
DSE7	2.239
DSE8	2.011
DSE9	1.930
ICT1	1.663
ICT10	1.603
ICT4	1.690

ICT5	2.128
ICT6	1.958
SDL2	3.381
SDL3	3.234
SDL4	2.159
SDL5	2.153

All indicator level VIF values fell within acceptable limits after removing collinear items, confirming no problematic redundancy among indicators within each construct. This means no single question was excessively duplicating information already captured by another question in the same construct, which keeps each item's contribution meaningful.

Collinearity for Variables: VIF Inner Model

	VIF
AIS -> DSE	1.418
AIS -> SDL	1.549
DSE -> SDL	1.950
ICT -> DSE	1.409
ICT -> SDL	1.935

All construct level VIF values remained well below the 5.0 threshold, confirming that multicollinearity among predictor constructs does not threaten the stability of the structural path estimates. In practical terms, this means ICT Preparedness and AI Support are independent enough from each other that the model can reliably separate their individual effects on the outcomes.

R-Square

	R-square
SDL	0.609

Sustainable Digital Leadership recorded an R^2 of 0.609, indicating that the model explains 60.9% of the variance in SDL a moderate to substantial level of explanatory power for behavioral research (Hair et al., 2022). This means the three predictors included in this study ICT Preparedness, AI Support, and Digital Self-Efficacy together account for a meaningful majority of what drives sustainable digital leadership behavior, though other unexamined factors likely contribute to the remaining variance.

Path Coefficients and Hypothesis Testing

Path	Path Coefficients	Standard Deviation	T statistics	P values	Bias		Result
					(Lower) 2.5%	(Upper) 97.5%	
ICT -> DSE	0.519	0.041	12.690	0.000	0.435	0.594	Supported
ICT -> SDL	0.105	0.048	2.192	0.028	0.011	0.198	Supported
AIS -> DSE	0.259	0.049	5.341	0.000	0.162	0.353	Supported
AIS -> SDL	0.512	0.038	13.568	0.000	0.442	0.590	Supported
DSE -> SDL	0.285	0.045	6.363	0.000	0.193	0.370	Supported
ICT -> DSE -> SDL	0.148	0.028	5.333	0.000	0.096	0.205	Supported
AIS -> DSE -> SDL	0.074	0.016	4.486	0.000	0.045	0.111	Supported

This table shows that all seven hypothesized paths were statistically significant, meaning every relationship proposed in the conceptual model was supported by the data. Employer Preparedness for ICT was the strongest predictor of Digital Self-Efficacy, AI Support was the strongest direct predictor of Sustainable Digital Leadership, and both indirect **(mediation)** paths through Digital Self-Efficacy were significant confirming that Digital Self-Efficacy plays a genuine bridging role rather than being a redundant variable in the model.

Conclusion and Suggestions

This study set out to understand what drives Sustainable Digital Leadership in Indian IT firms and how Digital Self-Efficacy fits within that process. The results indicate that organizations investing in ICT infrastructure and structured AI collaboration are building more than technical capacity they are cultivating the digital confidence that enables employees to think and act as sustainable digital leaders. AI Support stood out as the most influential factor, suggesting it is becoming central to leadership development in Indian IT organizations, while ICT preparedness provides the foundational conditions that make confident digital engagement possible. Digital Self-Efficacy sits at the center of this process, functioning as the mechanism through which organizational investment becomes individual leadership behavior.

Organizations should treat AI integration as a people development initiative rather than a purely technological rollout, with role specific guidance, proper onboarding, and visible leadership modeling of good AI use

ICT investment should be paired with ongoing training and accessible support, with HR teams mapping employees' digital development needs and building structured learning paths.

Because Digital Self-Efficacy functions as the critical bridge between organizational support and leadership outcomes, building employee digital confidence should be a deliberate organizational goal, supported through peer mentoring, lowstakes practice opportunities, and recognition of digital growth.

Progress should be tracked through regular employee surveys, digital capability assessments, and leadership audits, with sustainable digital leadership modeled visibly at senior levels and reinforced through performance systems.

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